UG (NEP) 4th SEMESTER
COMPUTER APPLICATIONS: COMPUTING MATHEMATICS
(CAP422J3)

Session: June 2025 Date of Examination: 24-06-2025
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 36
No. of Questions: 09

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

i. State one property of determinants.
ii. Define domain and range of a function.
iii. Define accuracy and precision.
iv. What is the derivative of sin(x)?
v. Give the domain of the function f(x) = 1/(x − 2).
vi. What is a one-to-one function?
vii. Define machine epsilon.
viii. Find the determinant of the matrix [[2, 3], [5, 7]].

Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Find the derivative of $f(x) = x^3 + 2x^2 - 5x + 7$.

OR

Differentiate $\sin(x)/(x^2 + 1)$ using the quotient rule.

3. Solve the system: 2x + 3y = 5, 4x + 5y = 6, using the matrix method.

OR

Add and multiply the matrices A = [[2, 3], [6, 3]] and B = [[0, 4], [3, 7]].

4. Find the second derivative of $y = \sin^2(x)$.

OR

Explain significant figures with examples.

5. Discuss floating-point representation with an example.

OR

Evaluate $\lim (x^2 + 16)/(x - 4)$, as x approaches 4.

Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain various types of matrices and matrix operations with examples.
7. Discuss limits, standard limits and evaluate $\lim (e^x - 1)/x$, as x approaches 0.
8. Find the second derivative of $f(x) = x^3 - 6x^2 + 9x + 15$ and find its critical points.
9. Explain round-off and truncation errors. How do errors propagate in numerical computation?